

Mark Dodici

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RESEARCH INTERESTS

Binary stars in dense clusters; transient phenomena in galactic nuclei; hierarchical triples and few-body dynamics; stellar tides; intermediate-mass black holes; celestial mechanics; galactic dynamics

EDUCATION

University of Toronto

2022 – (expected Aug 2027)

Ph.D. candidate, Department of Astronomy & Astrophysics (UofT DAA)

Graduate researcher, Canadian Institute for Theoretical Astrophysics (CITA)

Visiting Graduate Student, Institute for Advanced Study (Winter 2026)

Advisors: Professors Yanqin Wu & Scott Tremaine

Princeton University

2018 – 2022

A.B. Astrophysical Sciences, *magna cum laude*

Thesis advisors: Dr. Christopher Spalding & Professor Jeremy Goodman

PUBLICATIONS 4 peer-reviewed (2 first-author); 70 citations (33 first-author).

FIRST AUTHOR

5. **M. Dodici**, 2026. Binary disruptions driven by massive disks around massive black holes. [arXiv:2607.26205](https://arxiv.org/abs/2607.26205). Submitted to ApJ. 24 pp.
4. **M. Dodici**, S. Tremaine, & Y. Wu, 2026. Dynamical evolution of stellar binaries in galactic centers. [ApJ, 1000, 226](#). 13 pp.
3. **M. Dodici** & S. Tremaine, 2024. Studying binary formation under dynamical friction using Hill's Problem. [ApJ, 972, 193](#). 26 pp.

CONTRIBUTING

2. B. Hensley, C. Murray, & **M. Dodici**, 2022. Polycyclic Aromatic Hydrocarbons, Anomalous Microwave Emission, and their Connection to the Cold Neutral Medium. [ApJ, 929, 23](#). (9 pp.)

ON EDUCATION

1. B. Lewis et al. (incl. **M. Dodici**), 2025. Improving Undergraduate Astronomy Students' Skills with Research Literature via Accessible Summaries: A Case Study with Astrobites-based Lesson Plans. [Phys. Rev. Physics Education Research, 21, 010124](#). (17 pp.)

FELLOWSHIPS, GRANTS, and AWARDS

Jui Lin (Allen) Yen Award (UofT DAA, for outstanding student research)	2026
Lachlan Gilchrist Fellowship (UofT School of Graduate Studies)	2026
Duncombe Student Research Prize (AAS Division on Dynamical Astronomy)	2026
Carl Reinhardt Fellowship in Astronomy (UofT DAA)	2025
Dunlap Institute Seed Funding Grants for outreach workshop series	2023, 2024
Graduate Scholarship (CITA)	2023

TALKS 22 total; 6 invited (*), 1 award-based (†). Upcoming italicized.

AT CONFERENCES

22. * Galactic Nuclei Astrophysical Phenomena Sexten, Italy. Jul 2026
21. † AAS Division on Dynamical Astronomy 57 Chicago, IL. Jun 2026

20. CITA @ 40 / Kingston Meeting 2026	Toronto, ON. May 2026
19. Traversing the Galactic Center in Space and Time / IAU Symposium 405	Brno, Czechia. May 2026
18. MODEST-25 / IAU Symposium 398	Seoul, South Korea. Jun 2025
17. AAS Division on Dynamical Astronomy 56	Atlanta, GA. May 2025
16. AAS Division on Dynamical Astronomy 55	Toronto, ON. May 2024
15. AAS Division for Planetary Science 55	San Antonio, TX. Oct 2023
14. CITA Planet Day	Toronto, ON. Aug 2023
13. Emerging Researchers in Exoplanet Science Symposium VIII	New Haven, CT. Jun 2023
12. Great Lakes Exoplanets Area Meeting	Columbus, OH. Nov 2022
11. AAS 240	Pasadena, CA. Jun 2022

AT INSTITUTIONS

10. * <i>Astronomy Lunch Seminar, University of California, Los Angeles</i>	Los Angeles, CA. Dec 2026
9. <i>Theory Talk, Northwestern University/CIERA</i>	Evanston, IL. Nov 2026
8. * <i>Tea Talk, Indiana University</i>	Bloomington, IN. Nov 2026
7. * <i>APS Lunch Seminar, University of Colorado, Boulder</i>	Boulder, CO. Oct 2026
6. Galactic Center Group, Max Planck Institute for Extraterrestrial Physics	Garching, Germany. May 2026
5. Rossi Group, Leiden Observatory	Leiden, Netherlands. May 2026
4. * Dynamics Blackboard Talk, Institute for Advanced Study	Princeton, NJ. Mar 2026
3. * Astrophysics Seminar, Institute for Advanced Study	Princeton, NJ. Feb 2026
2. Compact Objects Group, CITA	Toronto, ON. Feb 2024
1. Lunch Talk, UofT DAA	Toronto, ON. Jan 2024, Apr 2026

LEADERSHIP and SERVICE ROLES

Referee, <i>ApJ, ApJL, OJA</i>	2025 –
Coding the Cosmos (HS workshop series), <i>Co-director (2024 –)</i>	2023 –
UofT Graduate Astronomy Students Association	2022 –
<i>Co-President (2024 – 2025)</i>	
<i>Courses & Qualifying Exams Committee (2022 – 2024, 2025 –)</i>	
Department Jamboree, <i>Organizer & Emcee</i>	2023 – 2025
UofT AstroTours (Public lecture/telescope events), <i>Co-director (2023 – 2024)</i>	2022 – 2024
Age of the Universe (HS workshop), <i>Co-organizer & Speaker</i>	2023

TEACHING ASSISTANT EXPERIENCE

Stars and Planets (AST221, Toronto)	Fall 2024, 2025, 2026
The Sun and its Neighbours (AST101, Toronto)	Fall 2022, 2023
<i>Lecturer, Summer 2026</i>	
Introduction to Astrophysics (AST320, Toronto)	Winter 2025
Stars and Galaxies (AST201, Toronto)	Winter 2023, 2024
The Universe (AST203, Princeton)	Winter 2021, 2022

OTHER OUTREACH

UofT Astrotours, <i>Speaker</i>	2026
Pursue STEM (HS science program), <i>Workshop leader</i>	2025
Astrobits (Research blog), <i>Author & Education Study co-author</i>	2023 – 2025
Eclipse 2024 at the Toronto Public Library (Public lecture series), <i>Speaker</i>	2024
Solar Eclipse 2024: Indigenous Knowledge (Video), <i>Editor</i>	2024
Cosmos from your Couch (Astronomy video series), <i>Editor</i>	2023
ComSciCon Canada (Science Communication Workshop), <i>Attendee</i>	2023
Princeton Research Day (Virtual talk competition), <i>Award-winning presenter</i>	2021